

Control of a chemical plant

Consider the chemical plant described in the attached paper (see Section 7: Example).

We define $x = [H_1, x_{A1}, x_{B1}, T_1, H_2, x_{A2}, x_{B2}, T_2, H_3, x_{A3}, x_{B3}, T_3]^T$ and $u = [F_{f1}, Q_1, F_{f2}, Q_2, F_R, Q_3]^T$. More specifically, the latter variables are defined as the differences with respect to the nominal equilibrium point around which we linearize. The “centralized” linearized model is

$$\dot{x} = Ax + Bu$$

where the system matrices are specified in the file. We assume that all state variables are measurable.

Problem:

1. Decompose the state and input vectors into subvectors, consistently with the physical description of the system. Obtain the corresponding decomposed model.
2. Generate the system matrices (both continuous-time and discrete-time, the latter with a sampling time $h = 0.1$ s). Perform the following analysis:
 - a. Compute the eigenvalues and the spectral abscissa of the (continuous-time) system. Is it open-loop asymptotically stable?
 - b. Compute the eigenvalues and the spectral radius of the (discrete-time) system. Is it open-loop asymptotically stable?
3. For different state-feedback control structures (i.e., centralized, decentralized, and different distributed schemes) perform the following actions
 - a. Compute the continuous-time fixed modes
 - b. Compute the discrete-time fixed modes
 - c. Compute, if possible, the CONTINUOUS-TIME control gains using LMIs to achieve the desired performances. Apply, for better comparison, different criteria for computing the control laws.
 - d. Compute, if possible, the DISCRETE-TIME control gains using LMIs to achieve the desired performances. Apply, for better comparison, different criteria for computing the control laws.
 - e. Analyze the properties of the so-obtained closed-loop systems (e.g., stability, eigenvalues) and compute the closed-loop system trajectories (generated both in continuous-time and in discrete-time) of the variables H_1 , H_2 , and H_3 starting from a common random initial condition.